Enhanced Electron Nuclear control leveraging hyperfine coupling and radio-frequency drive

Donghun Jung[1, 2], Eunsang Lee[1], Jiwon Jeon[1], Junghyun Lee[1]

[1]Center for Quantum Information, Korea Institute of Science and Technology, Seoul 02792, Republic of Korea
[2]Department of Physics, Sungkyunkwan University, Suwon 16419, Republic of Korea

Abstract

Solid-state defect systems have emerged as promising platforms for quantum information processing and sensing. In particular, the nitrogen-vacancy (NV) center in diamond, coupled with surrounding nuclear spins, exhibits long coherence times and enables high-fidelity operations. To implement high-fidelity, selective-qubit operations, dynamic decoupling sequences, consisting of multi-pulse control on the NV spin, have been developed to achieve entangling gates with selected target nuclear spins while reducing decoherence from the broader spin bath. On the other hand, an additional selective phase-controlled radio-frequency (rf) driving on nuclear spins, referred as DDrf gate, enables elaborated control of individual spins. The mechanisms underlying these approaches differ that dynamic decoupling relies on transverse hyperfine interaction components, whereas the DDrf gate exploits rf driving and suppresses transverse hyperfine coupling. In this work, we propose a hybrid DDrf gate that leverages both mechanisms to achieve faster and more robust gate operations. This approach could enable improved quantum control and sensing capabilities.

Conditional Gate

Beyond a mere collection of independent qubits, entanglement between the qubits is regarded as a core resourse for quantum speed up, ... Two-qubit gates is the source of entanglement between qubit, so implementing two-qubit gate in a particular physical quantum system is critical interest. Generally such operation can be implemented by conditional operation, that is, a two-qubit gate acts on a control qubit and a target qubit. The operation applied to the target depends on the state of the control, its operation can be written as:

$$U = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1,$$

where U_0 and U_1 are single-qubit rotations conditioned on the control qubit state. Without loss of generality, we can write each operaion as $e^{-i\theta_i \vec{\sigma}_i}$. When the rotating axes $\vec{\sigma}$ on the target qubit are significantly different, or even antiparallel, the gate operation becomes conditional, ultimately generate entanglement.

Nitrogen-Vacancy (NV) center system

In the nitrogen-vacancy (NV) center, such operation can be implemented by a sort of Hamiltonian engineering. We consider a nitrogen-vacancy (NV) center electron spin ($S = 1$) coupled to surrounding ^{13}C nuclear spins ($I = \frac{1}{2}$) via the hyperfine interaction [?]. The Hamiltonian is

$$H = \omega_0 I_z + A_\parallel S_z I_z + A_\perp S_z I_x,$$

where ω_L is the nuclear Larmor frequency, and $A_\parallel$, $A_\perp$ are the parallel and perpendicular hyperfine components. We can rewrite Hamiltonian as:

$$H = |0\rangle\langle 0| \otimes \underbrace{\omega_L I_z}_{=H_0} + |1\rangle\langle 1| \otimes \underbrace{(\omega_L + A_\parallel)\, I_z + A_\perp\, I_x}_{=H_1}$$

We can evaluate time evolution of ^{13}C nuclear spin depending on NV center electron spin state in a piecewise manner.

Dynamic Decoupling Based Conditional Gate

We can enable conditional interaction by applying a dynamic decoupling sequence consisting of N sequencial π-pulses. Consider the basic decoupling unit on the electron spin τ-π-2τ-π-τ where τ is a free evolution time. The net result of this unit sequence results in rotaion by an angle ϕ around an axis $\vec{\sigma}_i$, depending on the initial state of electron spin; $\vec{\sigma}_0$ for $m_s = 0$ state, and $\vec{\sigma}_1$ for $m_s = -1$ state. If $\vec{\sigma}_0$ and $\vec{\sigma}_1$ are not parallel, the resulting conditional rotation of the nuclear spin generally entangles the electron and nuclear spins.

CPMG Pulse Sequence

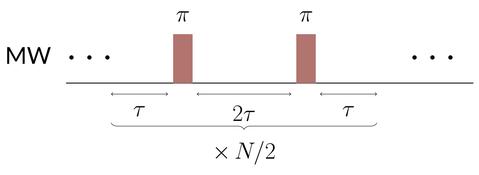

Figure 1. CPMG dynamical decoupling sequence on the NV electron spin.

A dynamical decoupling sequence with the basic unit τ-π-2τ-π-τ, repeated $N/2$ times, makes further conditional operation.

Conditional Rotation & Resonance

Let V_0 and V_1 are the conditional evolution operators for a single decoupling unit:

$$V_0 = e^{-i\tau H_0} e^{-i2\tau H_1} e^{-i\tau H_0} \qquad = e^{-i\phi\vec{\sigma}_0} \tag{1}$$
$$V_1 = e^{-i\tau H_1} e^{-i2\tau H_2} e^{-i\tau H_1} \qquad = e^{-i\phi\vec{\sigma}_1} \tag{2}$$

The conditional rotation angle ϕ is determined by the choice of τ and NV center system configuration.

$$\cos\phi = \cos\omega_1\tau\cos\omega_0\tau - \frac{\omega_0 - A_\parallel}{\omega_1}\sin\omega_1\tau\sin\omega_0\tau \tag{3}$$

We might investigate the inner product between two rotation axes.

$$1 - \vec{\sigma}_0 \cdot \vec{\sigma}_1 = \frac{A_\perp^2}{\omega_0^2}\frac{(1-\cos\omega_1\tau)(1-\cos\omega_0\tau)}{1+\cos\phi} \tag{4}$$

That is, τ must be tuned to the specific resonance condition, and the gate speed is contingent on the transverse hyperfine component $A_\perp$. The resonant τ satisfies

$$\tau_k = \frac{(2k+1)\pi}{\omega_0 + \omega_1}. \tag{5}$$

DDrf Gate

The DDrf gate combines dynamical decoupling on the electron spin with phase-controlled radio-frequency (rf) driving of the nuclear spin [?]. An rf field at frequency $\omega_1 = \sqrt{(\omega_L - A_\parallel)^2 + A_\perp^2}$ (the nuclear precession frequency for $m_s = -1$) selectively drives the nuclear spin depending on the electron state.

Hamiltonian

In the interaction picture with respect to the electron energy splitting and neglecting non-secular terms, the Hamiltonian of the NV-^{13}C system is given by

$$H = \omega_0 I_z + A_\parallel S_z I_z + A_\perp S_z I_x + 2\Omega_{\text{RF}}(\omega t + \phi) I_x \tag{6}$$

where the final term describes the interaction of the nuclear spin with a radio frequency (RF) driving field polarized along x axis (on the exact same direction to hyperfine coupling), phase ϕ and Rabi frequency Ω_{RF}.

In the rotating frame at the rf frequency, and approximation $A_\perp \approx 0$ and $\Omega_{\text{RF}} \ll (\omega_0 - \omega_1)$ for simplicity:

$$H = |0\rangle\langle 0| \otimes \underbrace{(\omega_0 - \omega_1) I_z}_{=H_0} + |1\rangle\langle 1| \otimes \underbrace{\Omega_{\text{RF}}(\cos\phi I_x + \sin\phi I_y)}_{=H_1}. \tag{7}$$

where Ω is the rf Rabi frequency and ϕ is the rf phase. For $m_s = 0$ the nuclear spin precesses freely; for $m_s = -1$ it is driven by the rf field.

Pulse Sequence

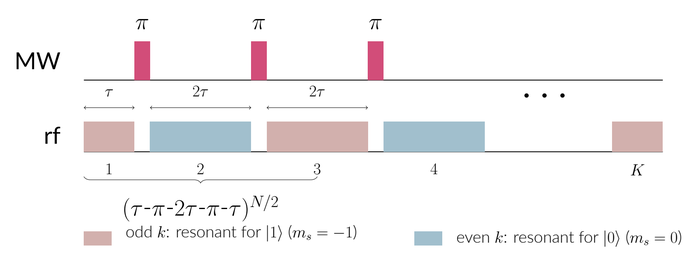

Figure 2. DDrf pulse sequence. DD π-pulses on the electron (MW) are interleaved with phase-controlled rf pulses ($k = 1, \ldots, K$; $K = N + 1$) on the nuclear spin.

The electron spin state determines which rf pulses are resonant: odd-k pulses drive the nuclear spin when the electron is in $|1\rangle$ ($m_s = -1$), while even-k pulses are resonant for $|0\rangle$ ($m_s = 0$).

Phase Prescription

The rf pulse phases $\phi_k = \varphi + \phi'_k$ are set to accumulate constructive rotations:

$$\phi'_k = \begin{cases} (k-1)\phi_\tau + \pi, & k \text{ odd} \\ (k-1)\phi_\tau, & k \text{ even} \end{cases}, \qquad \phi_\tau = A_\parallel \tau.$$

Controlled Rotation

The resulting two-qubit gate is a conditional rotation for each unit:

$$U = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1,$$

where

$$U_0 = R_z(N(\omega_0 - \omega_1)\tau) R_\varphi(N\Omega_{\text{rf}}\tau) \tag{8}$$
$$U_1 = R_z(N(\omega_0 - \omega_1)\tau) R_\varphi(-N\Omega_{\text{rf}}\tau) \tag{9}$$

where $R_\varphi(\theta) = e^{-i\theta[\cos\varphi I_x + \sin\varphi I_y]}$.

Hybrid DDrf Gate

We propose a hybrid gate that combines both mechanisms. The transverse hyperfine interaction ($A_\perp$) exploited by the DD gate and the rf driving (Ω) exploited by the DDrf gate. By allowing both contributions to act simultaneously, the hybrid approach can achieve:

- **Faster gate operations** through constructive combination of both driving mechanisms.
- **availability** that achieve high fidelity without dependence on spin configuration.

In order to keep conditional operation coming from the $A_\perp$ and the choice of τ, we would like to choose τ such that $\vec{\sigma}_0 \cdot \vec{\sigma}_1 \simeq -1$ or $\tau = \frac{(2k+1)\pi}{\omega_0 + \omega_1}$ for a certain k. Then, over conditional operation by dynamic decoupling, the speed of conditional operation is boosted by additional rf driving. In the region of small $A_\perp$ where $\sin\beta \simeq \beta$, effectively we can combine each conditional operation addictively. That is, conditional angle $\phi\phi_{DD} + \phi_{DDrf}$. That is, with chosen initial phase $\varphi = 0$,

$$U = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1, \tag{10}$$

where

$$U_0 = R_x(N(\beta\sin(\omega_0\tau) + \Omega_{\text{rf}}\tau)) \tag{11}$$
$$U_1 = R_x(-N(\beta\sin(\omega_0\tau) + \Omega_{\text{rf}}\tau)) \tag{12}$$

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